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pi@raspberrypi ~ $ uname -a
Linux raspberrypi 4.1.6-v7+ #810 SMP PREEMPT Tue Aug 18 15:32:12 BST 2015 armv7l
GNU/Linux

pi@raspberrypi ~ $ sudo apt-get install xrdp
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  tightvncserver xfntt-base
Suggested packages:
  tightvnc-java
The following NEW packages will be installed:
  tightvncserver xfntt-base xrdp
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 7,219 kB of archives.
After this operation, 11.5 MB of additional disk space will be used.
Do you want to continue [Y/n]?

```

Fig. 8: Installing XRDP for remote computing

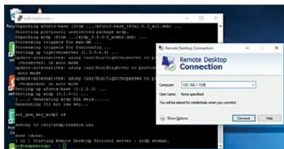


Fig. 9: Entering IP address



Fig. 10: Entering username and password

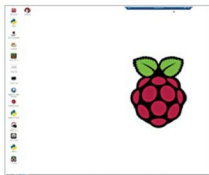


Fig. 11: RPi desktop

Bluetooth module. HC-05 module is an easy-to-use Bluetooth SPP (serial port protocol) module designed for transparent wireless serial connection setup. Serial port Bluetooth module is a fully qualified Bluetooth V2.0 + EDR (Enhanced Data Rate) 3Mbps modulator with complete 2.4GHz radio transceiver and baseband. It uses CSR

Bluecore 04-External single-chip Bluetooth system with CMOS technology and AFH (adaptive frequency hopping) feature. Its default baud rate is 38400, has eight data bits, one stop bit and no parity. It supports following baud rates: 9600, 19200, 38400, 57600, 115200, 230400 and 460800.

L293D motor driver. L293D is a typical motor driver IC that allows DC motor to drive in either direction. L293D is a 16-pin IC that can control a set of two DC motors simultaneously in any direction. It means you can control two DC motors with a single L293D IC.

Camera module. A camera module is an image sensor integrated with a lens, control electronics and an interface like CSI, Ethernet or plain raw low-voltage differential signaling. The Raspberry Pi camera module can be used to take high-definition videos as well as still photographs. The module has a five megapixel fixed-focus camera that supports 1080p30, 720p60 and VGA90 video modes as well as stills capture. It attaches via a 15cm ribbon cable to the CSI port on the Raspberry Pi. It can be accessed through the MMAL and V4L APIs, and there are numerous third-party libraries built for it, including the Picamera Python library.

Raspberry Pi and video configurations. RPi is used in this project because it allows easy video transmission over Wi-Fi. The RPi is considered as the CPU of the robot. The Raspberry Pi 3 (RPi 3) Model B used here is the third-generation RPi. This powerful

credit-card sized single-board computer can be used for many applications. RPi 3 Model B released in February 2016 is bundled with on-board Wi-Fi, Bluetooth and USB boot capabilities. The RPi 3 uses a Broadcom BCM2837 SoC with a 1.2GHz 64-bit quad-core ARM Cortex-A53 processor, with 512kB shared L2 cache.

The next main step is the setting up of the RPi. After initial configuration of RPi is done, the RPi board is connected to Wi-Fi. An IP address is programmed and setup is linked to the RPi. The video captured by the RPi camera is sent over the Wi-Fi modem. This video can be viewed in your smartphone by connecting to the same Wi-Fi connection and IP address of the RPi.

The RPi is powered up using a 5V, 1000mAh power bank (Xiaomi). Then an SD card with the Raspbian software is inserted into the slot provided in the RPi board. The board is connected to a monitor for further configuration. The RPi configuration tool in Raspbian allows you to easily enable features and change your specific settings like keyboard layout. The procedure for the initial setting up is as follows:

1. Access the Terminal window on the Mac (or equivalent on another machine)
2. Run wirelessly with a static IP for each SD card
3. Automatically startup
4. Set the time zone
5. Skip the GUI of the RPi

Once the above setup is complete, we can see the desktop screen on RPi monitor. In order to view the desktop again we need to access via the remote desktop.

To get the remote desktop in some laptops it is mandatory to install the XRDP software, as shown in Fig. 8. The user name and password are entered in the screen. Then IP address of the computer is entered as shown in Fig. 9. The RPi is accessed by entering the username and password as shown in Fig. 10. The desktop of the RPi opens up as shown in Fig. 11.

Software

Apache web server. Apache is used in this project to configure RPi as server. Apache is a popular web server appli-